

Clifford Alg : In D -dim $n=1, \dots, D$

$$\{\gamma_m, \gamma_n\} = \gamma_m \gamma_n + \gamma_n \gamma_m = 2\eta_{mn} \mathbb{1}$$

η_{mn} is the metric on $\mathbb{R}^{s,t}$, $s+t=D$

$$\eta_{nm} = \text{diag}(\underbrace{-1, -1, \dots, -1}_s, \underbrace{1, \dots, 1}_t)$$

* The elements γ_m and $\mathbb{1}$ generate the finite group

$$C_D \equiv \{ \pm \mathbb{1}, \pm \gamma_m, \pm \gamma_{m_1 m_2}, \dots, \pm \gamma_{n_1 \dots n_p} \}$$

$$|C_D| = 2 \sum_{p=0}^D \binom{D}{p} \quad \text{2: from } \pm \text{ of each}$$

$$(x+y)^n = \sum_{p=0}^n \binom{n}{p} x^p y^{n-p}$$

$$\text{Since we have } \sum_{p=0}^D \binom{D}{p} \Leftrightarrow x=y=1$$

$$\therefore (x+y)^n = (1+1)^n = \sum_{p=0}^D \binom{D}{p} = 2^D$$

$$|C_D| = 2 \cdot 2^D = 2^{D+1}$$

Cliff Alg. in Even dim:

* The number of irreducible reps of any finite dim group G equals the number of its conjugacy classes $a \in G$

$$\left[\begin{array}{l} \text{Conj. Class} \equiv [a] := \{g a g^{-1} \mid \forall g \in G\} \\ a \sim b \Leftrightarrow \exists g \in G \text{ s.t.} \\ a = g b g^{-1} \end{array} \right.$$

$$\{\gamma_m, \gamma_n\} = 2\eta_{mn} \mathbb{1}$$

Conj. Classes of C_D :

$$[+1] : 1 = g b g^{-1} \Rightarrow b = 1 \quad [+1] \approx \{1\}$$

$$[-1] \equiv \{-1\}$$

$$[\gamma_m] :$$

$$[\gamma_{m_1 \dots m_p}] :$$

$[m_1, \dots, m_0]$:

* There is a separate conj. class for each element of $\gamma_{m_1, \dots, m_0}$

Thus, $\binom{p}{p}$ conj. classes for $\gamma_{m_1, \dots, m_p}$

$p=0$: $\gamma_{1_0} = 1$, single element:

$$\sum_{p=0}^D \left(\sum_{p=0}^D \binom{p}{p} + 1 \right) = 2^D + 1$$

* Thus, there are $2^D + 1$ inequivalent irreps of C_D

Claim: (# of inequivalent 1D reps of G)
 $= |G| / |\text{Com}(G)|$

Where $\text{Com}(G)$ is the commutator (Center) of G defined by

$$\text{Com}(G) = aba^{-1}b^{-1}, \quad \forall a, b \in G$$

* $\text{Com}(C_D) = \{\pm 1\}$

We note $\text{Com}(C_D) = \gamma^{(\alpha)} \gamma^{(\beta)} [\gamma^{(\alpha)}]^{-1} [\gamma^{(\beta)}]^{-1}$
 to commute $\gamma^{(\alpha)} \gamma^{(\beta)}$ for $\alpha, \beta \in \{m_1, \dots, m_p\}$
 $\gamma^{(\alpha)} \gamma^{(\beta)} = \pm 1 \gamma^{(\beta)} \gamma^{(\alpha)}$

and so in the end we are left with

$$\text{Com}(C_D) = \pm 1$$

$$\therefore |\text{Com}(C_D)| = 2$$

$$(\# \text{ of inequivalent 1D reps}) = \frac{|C_D|}{2}$$

$$= (2^{p+1})/2$$

$$= 2^p$$

* Thus: There is only one irrep of C_D whose $\dim \geq 1$

$\leadsto$ p inequivalent irreps: ρ_i
 $|\rho_i| \equiv n_p$

Then $|G| = \sum_p |\rho_p|^2$

ex: $C_D: 2^{p+1} = n_1^2 + \dots + n^2$
 $= 1^2 + 1^2 + \dots + n^2$
 $= 1^2(2^p) + n^2$

2^p since there are $2^p + 1$ and the 1 is for the only non 1D rep.

$$2^{p+1} = 2^p + n^2$$

$$2^p(2 - 1) = n^2$$

$$n = 2^{p/2}$$

$$\therefore \dim(\chi^{\text{irrep}}) = n = 2^{p/2}$$

Thm: For $D = 2m, m \in \mathbb{Z}$ the group C_D has $2^p + 1$ ineq. irreps. 2^p are 1D and the remaining rep is of $\dim 2^{p/2}$

Suppose the $\dim$ of an irrep is $\tilde{n}$, this means

$$\tilde{\rho}: G \rightarrow V^{\tilde{n}}$$

$$\dim(V^{\tilde{n}}) = \tilde{n}$$

As such, as a matrix rep

$\phi^\alpha_\beta V^\beta, \alpha, \beta = 1, \dots, \tilde{n}$
Thus, we can represent Y_m as $\tilde{n} \times \tilde{n} = 2^{p/2} \times 2^{p/2}$
matrices.